Let be the rod's angle to the horizontal when ball leaves (see figure). Then

(or rad).

Scoring (points): ; ; each ; ; ;
; ; each ; each .

IV. (points)

Method

Let be 's mass, the angle of to the vertical, and its angular speed. For pair , repulsion is . Newton's law gives

Using in gives

The geometry gives

With the given expressions for and , gives

Also, give

Substituting the given expressions for and gives